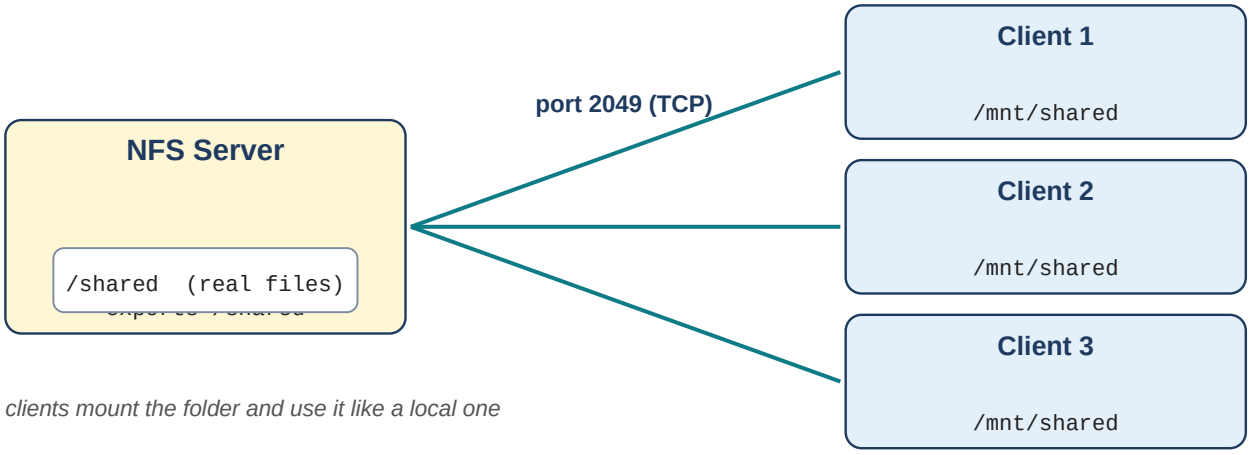
NFS and Amazon EFS

Network File System in simple words: how it works, how AWS EFS uses it, and how to create and mount it with commands

Part	Topic	One-line idea
1	What is NFS?	Share one folder with many computers
2	Why NFS, features and versions	Central storage, port 2049
3	NFS in AWS: Amazon EFS	Managed NFS: regional and multi-AZ
4	Create EFS (steps)	Console walk-through
5	Mount EFS on EC2	Commands to copy and paste
6	Build your own NFS server	The protocol itself, on two instances
7	Troubleshooting	Why will it not mount?
8	Quick quiz	Revise with answers

1. What is NFS?

NFS stands for **Network File System**. It is a way for computers to **share files over a network**. You keep one storage location on one machine, and many other computers connect to it and use it **as if it were their own local disk**. It is like **Google Drive or OneDrive, but for servers inside a network**.



clients mount the folder and use it like a local one

Part	Meaning
NFS server	The machine that has the real storage and **shares (exports)** a folder.
NFS client	Any machine that connects to the server and uses the shared folder.
Export	A folder the server offers to others.
Mount	Attaching the shared folder to a local folder, for example /mnt/shared. After mounting, you read and write files in it like any normal folder.

Easy picture: Office filing cabinet: the NFS server is the cabinet, and the employees (clients) do not keep their own copies. They open the same drawer and work on the same files. If one person updates a document, everyone sees the update straight away.

2. Why NFS? Features and Versions

	Without NFS	With NFS
Files	Each server has its own local copy	All servers point to the same storage
Sharing and syncing	Hard: copy files around	Automatic: one copy
Adding servers	Copy all data again	Just mount the shared folder

Key features

Feature	Meaning
Centralized storage	Files are in one place, not scattered over many machines.
Shared access	Many servers use the same data at the same time.
Scalability	Add more clients without copying files everywhere.
Transparency	To the client it looks like a normal folder.
Port 2049	The default port for NFS (TCP, and UDP in older versions). It must be open in the firewall or security group.

Use cases

- A website on **several servers**: all use the same images, code or logs from one NFS share.
- Shared home folders for users, shared data for analysis jobs and containers.
- In AWS, **EFS (Elastic File System)** is Amazon's managed NFS service.

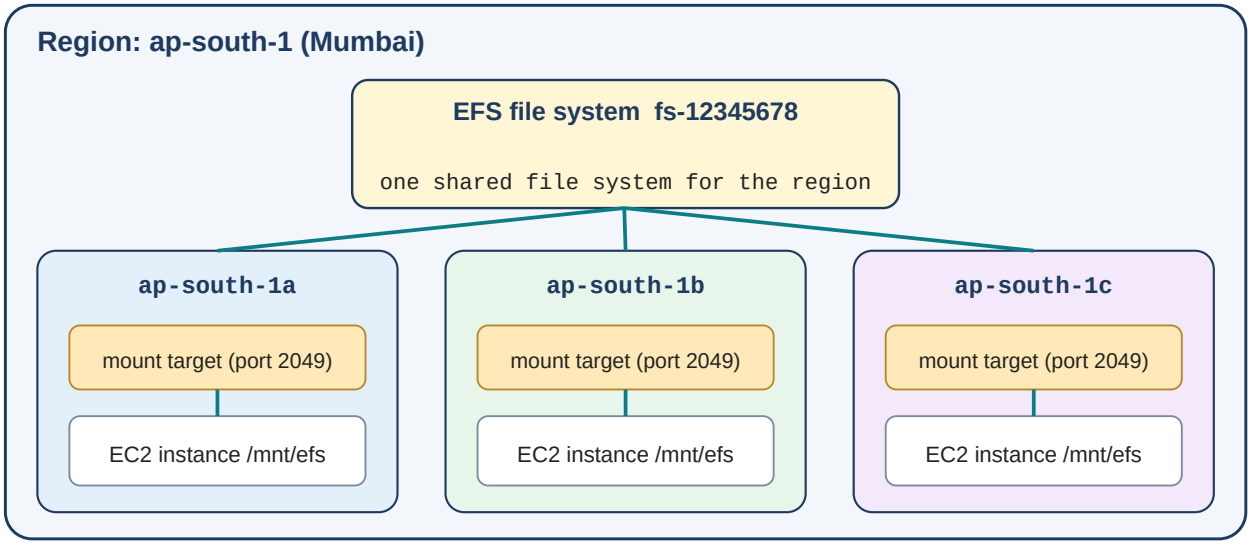
NFS versions

Version	Notes
NFSv2	Old and basic. Not used any more.
NFSv3	Supports large files and better performance. Needs extra helper services and ports.
NFSv4	Newer and more secure (ACLs, encryption options, stronger authentication). Works over one port, 2049.
NFSv4.1	The version AWS EFS uses by default (and 4.0 is also supported).

3. NFS in AWS: Amazon EFS

Amazon EFS (Elastic File System) is a **managed NFS file system**. AWS runs the servers; you just create it and mount it. It **grows and shrinks automatically** as you add or delete files, and you pay for the storage you use.

Question	Answer
Is the NFS protocol region or AZ specific?	No. It only needs a network path between client and server.
Is EFS region specific?	**Yes.** A file system exists in one AWS region.
Is EFS multi-AZ?	**Yes.** By default (Standard storage) data is stored across multiple AZs of that region for high availability.
Can instances in different AZs share it?	Yes, if they are in the same region. You create a **mount target** in each AZ.



Summary: EFS is region-specific, but it can be accessed from multiple AZs within that region. (There is also an EFS **One Zone** option that keeps data in a single AZ at a lower price. Use the regional Standard type for high availability.)

EFS vs EBS vs S3

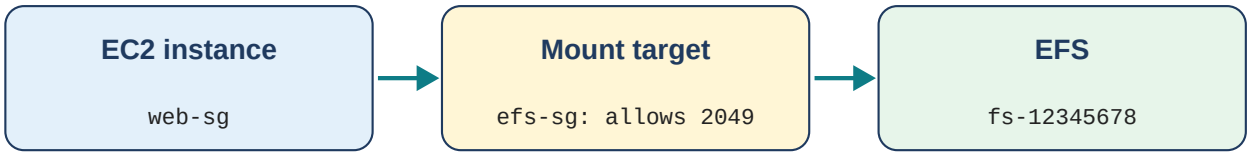
	EBS	EFS	S3
Type	Block storage (a disk)	File storage (NFS)	Object storage (web API)
Attached to	One instance (same AZ)	Many instances, many AZs	Not attached: accessed by API or URL
Size	You choose it, grow manually	Grows and shrinks automatically	Unlimited
OS	Linux and Windows	**Linux only**	Any (via API)
Typical use	Boot disk, database	Shared files, web content	Backups, media, static sites

EFS settings you will see

Setting	Meaning
Performance mode	General Purpose (default, recommended) or Max I/O (very high parallel workloads).
Throughput mode	Elastic (default: scales automatically), Provisioned or Bursting.
Storage class	Standard, and cheaper Infrequent Access / Archive for files you rarely use (via lifecycle management).
Encryption	At rest (choose when creating) and in transit (mount with the TLS option). Recommended.
Mount target	The network endpoint of EFS in one AZ/subnet, protected by a security group.

4. Step 1: Create the EFS File System

1. **Open Amazon EFS**
   AWS Console, search EFS, check the region, click Create file system
2. **Name and VPC**
   Name: my-efs. Select your VPC (the same VPC as your EC2 instances)
3. **Customize settings**
   Storage class Regional. Performance General Purpose. Throughput Elastic
4. **Encryption**
   Keep Encryption enabled (recommended)
5. **Network: mount targets**
   One in each AZ and subnet you use, for high availability
6. **Security group for the mount targets**
   Allow inbound NFS: TCP 2049, source = the security group of your EC2 instances
7. **Create**
   Click Create. Note the File system ID, for example fs-12345678



Remember: The most common mistake is the **security group**. The **EFS mount target** must allow **NFS (TCP 2049)** from the instances that mount it. Using the instance security group as the source is neater than IP addresses.

5. Step 2 and 3: Install the Client and Mount EFS

Install the NFS client tools on the EC2 instance

Operating system	Package	Command
Ubuntu / Debian	nfs-common	sudo apt-get update then sudo apt-get install -y nfs-common
Amazon Linux / RHEL / CentOS	amazon-efs-utils, nfs-utils	sudo yum install -y amazon-efs-utils nfs-utils

COPY AND PASTE: Ubuntu: install the NFS client

```
sudo apt-get update
sudo apt-get install -y nfs-common
```

COPY AND PASTE: Amazon Linux / RHEL: install the client

```
sudo yum install -y amazon-efs-utils
sudo yum install -y nfs-utils
```

Mount the file system

Syntax	Meaning
mount -t nfs4 -o OPTIONS SERVER:/ MOUNTPOINT	Mount an NFSv4 share. SERVER:/ means the root of the share.
mount -t efs FS_ID:/ MOUNTPOINT	Mount with the EFS helper (Amazon Linux). Add -o tls to encrypt in transit.
df -h / mount | grep nfs	Check that it is mounted.

COPY AND PASTE: Mount EFS with NFS (replace the file system ID and region)

```
FS=fs-12345678.efs.ap-south-1.amazonaws.com
OPTS=nfsvers=4.1,rsize=1048576,wsize=1048576
OPTS=$OPTS,hard,timeo=600,retrans=2,noresvport
sudo mkdir -p /mnt/efs
sudo mount -t nfs4 -o $OPTS $FS:/ /mnt/efs
df -h /mnt/efs
```

Command	What it does (in simple words)
FS=fs-12345678.efs.ap-south-1.amazonaws.com	The DNS name of your file system: fs-ID.efs.REGION.amazonaws.com. Use your own ID and region.
OPTS=...	The mount options (explained below), built in two lines to keep them readable.
sudo mkdir -p /mnt/efs	Creates the local folder (mount point).

Command	What it does (in simple words)
`sudo mount -t nfs4 -o $OPTS $FS:/ /mnt/efs`	Mounts the whole file system over NFSv4.1.
`df -h /mnt/efs`	Shows it mounted. EFS reports a huge size (8.0E) because it grows automatically.

Output (example)

```
Filesystem                                Size Used Avail Use% Mounted on
fs-12345678.efs.ap-south-1.amazonaws.com:/  8.0E    0  8.0E   0% /mnt/efs
```

Option	Meaning
nfsvers=4.1	Use NFS version 4.1 (the version EFS supports).
rsize / wsize = 1048576	Read and write in 1 MiB chunks (good performance).
hard	If EFS is not reachable, keep retrying instead of failing with an error (protects data).
timeo=600, retrans=2	Wait 60 seconds before retrying a request, retry twice.
noresvport	Use a new network port after a reconnect so that the connection recovers.

The short form from your notes also works for a quick test: `sudo mount -t nfs4 -o nfsvers=4.1 fs-12345678.efs.ap-south-1.amazonaws.com:/ /mnt/efs`. The longer options are the AWS recommendation.

Amazon Linux with the EFS helper (and encryption in transit)

COPY AND PASTE: Mount with amazon-efs-utils

```
sudo mkdir -p /mnt/efs
sudo mount -t efs -o tls fs-12345678:/ /mnt/efs
df -h /mnt/efs
```

Make the mount permanent (survives reboot)

The commands `systemctl daemon-reload` and `mount -a` in your notes are used after you add a line to **/etc/fstab**, the file that lists disks to mount at boot.

COPY AND PASTE: Add EFS to /etc/fstab

```
FS=fs-12345678.efs.ap-south-1.amazonaws.com
LINE="$FS:/ /mnt/efs nfs4 nfsvers=4.1,_netdev,nofail 0 0"
echo "$LINE" | sudo tee -a /etc/fstab
sudo systemctl daemon-reload
sudo mount -a
df -h /mnt/efs
```

Command	What it does (in simple words)
`LINE=... ; echo "$LINE" | sudo tee -a /etc/fstab`	Builds the line and adds it to fstab. _netdev waits for the network. nofail lets the server boot even if EFS is not reachable.
`sudo systemctl daemon-reload`	Makes systemd read the changed fstab.
`sudo mount -a`	Mounts everything in fstab. No error means the line is correct.
`df -h /mnt/efs`	Verify.

Careful: If you made a mistake in fstab, a wrong line can slow or stop the boot. That is why **nofail** matters, and why you should run **sudo mount -a** to test before rebooting.

Prove that the file system is shared

Mount the **same EFS** on **two instances**, ideally in different AZs. Create a file on the first and read it on the second.

COPY AND PASTE: On instance A, create a file

```
sudo chown $USER:$USER /mnt/efs
echo "Hello from instance A" > /mnt/efs/hello.txt
ls -l /mnt/efs
```

COPY AND PASTE: On instance B (another AZ), read it

```
cat /mnt/efs/hello.txt
```

Output (example)

```
-rw-r--r-- 1 ubuntu ubuntu 22 Dec 18 10:05 hello.txt     <-- on A
Hello from instance A                                       <-- on B: same file
```

Command	What it does (in simple words)
`sudo chown $USER:$USER /mnt/efs`	A new EFS root folder is owned by root. This gives your user permission to write.
`echo ... > /mnt/efs/hello.txt`	Writes a file into the shared folder.
`cat /mnt/efs/hello.txt`	On the other instance, the same file appears at once.

Unmount

COPY AND PASTE: Unmount when you are done

```
cd ~
sudo umount /mnt/efs
df -h | grep efs
```

6. Build Your Own NFS Server (the Protocol Itself)

EFS hides the server. To see how **NFS itself** works, turn one Ubuntu instance into an NFS server and mount it from a second. Both instances should be in the same VPC. In the **server's security group**, allow **TCP 2049** from the client's security group or private IP.

On the NFS server

COPY AND PASTE: Set up the NFS server (Ubuntu)

```
sudo apt-get update
sudo apt-get install -y nfs-kernel-server
sudo mkdir -p /srv/share
sudo chmod 777 /srv/share
LINE="/srv/share 172.31.0.0/16(rw,sync,no_subtree_check)"
echo "$LINE" | sudo tee -a /etc/exports
sudo exportfs -ra
sudo exportfs -v
sudo systemctl enable --now nfs-kernel-server
```

Command	What it does (in simple words)
`sudo apt-get install -y nfs-kernel-server`	Installs the NFS server software.
`sudo mkdir -p /srv/share ; sudo chmod 777 /srv/share`	The folder to share. chmod 777 lets everyone write: for practice only.
`LINE="/srv/share 172.31.0.0/16(...)" ; echo "$LINE" | sudo tee -a /etc/exports`	Adds the export rule: folder, allowed network, options. Use your VPC range, never *.
`sudo exportfs -ra`	Re-reads /etc/exports and applies it.
`sudo exportfs -v`	Lists what is being shared.
`sudo systemctl enable --now nfs-kernel-server`	Starts the service now and at every boot.

Export option	Meaning
rw	Clients can read and write (ro = read only).
sync	The server writes data to disk before it replies (safer).
no_subtree_check	Avoids a slow, rarely useful check. Recommended.
root_squash (default)	A root user on the client is treated as a normal user on the server (safer).

On the NFS client

COPY AND PASTE: Mount the share (replace SERVER_PRIVATE_IP)

```
sudo apt-get install -y nfs-common
SERVER=SERVER_PRIVATE_IP
sudo mkdir -p /mnt/share
sudo mount -t nfs4 $SERVER:/srv/share /mnt/share
df -h /mnt/share
echo "written by client" > /mnt/share/client.txt
```

Output (example)

```
Filesystem                Size  Used Avail Use% Mounted on
172.31.5.20:/srv/share  8.0G  1.9G  6.1G  24% /mnt/share
```

Back on the **server**, run `ls /srv/share`: **client.txt** is there. The file was written over the network, through port 2049.

7. Troubleshooting

Symptom	Likely cause and fix
mount hangs or times out	Security group of the mount target (or NFS server) does not allow TCP 2049 from the client. Or there is no mount target in the client's AZ.
Name or service not known	Wrong file system ID or region in the DNS name, or VPC DNS resolution is off.
wrong fs type, bad option, bad superblock	The NFS client tools are not installed: install nfs-common or nfs-utils.
access denied by server	The export does not list your client's IP (NFS server), or an EFS file system policy denies you.
Permission denied when writing	The folder is owned by root. Use chown or chmod on the share.
Boot is slow after adding fstab line	Add _netdev,nofail and test with sudo mount -a.

COPY AND PASTE: Quick checks from the client

```
FS=fs-12345678.efs.ap-south-1.amazonaws.com
nslookup $FS
timeout 3 bash -c "</dev/tcp/$FS/2049" 2>/dev/null && echo open || echo closed
mount | grep nfs
```

Command	What it does (in simple words)
`nslookup $FS`	Should return a private IP of the mount target in your VPC.
`timeout 3 bash -c "</dev/tcp/$FS/2049" ...`	Tests port 2049: open means the security group allows it.
`mount | grep nfs`	Shows active NFS mounts.

Cost tip: EFS charges for the data you store. When practice is over, **unmount**, then **delete the file system** (EFS, select it, Delete). Delete the practice instances and extra security groups too.

8. Quick Quiz and Summary

Question	Answer
What does NFS stand for?	Network File System.
What are the two parts of NFS?	NFS server (shares storage) and NFS clients (mount it).
Which port does NFS use?	2049.
Which NFS version does EFS use by default?	NFSv4.1.
Is the NFS protocol tied to a region or AZ?	No. It only needs network connectivity.
Is EFS regional or zonal?	Regional (Standard storage), stored across multiple AZs of one region.
Can instances in different AZs mount the same EFS?	Yes, in the same region, through a mount target in each AZ.
Which operating systems can mount EFS?	Linux (not Windows).
What must the mount target security group allow?	Inbound TCP 2049 from the instances.
Which file is edited to mount at boot?	/etc/fstab (then sudo mount -a to test).
EBS or EFS for files shared by many servers?	EFS.

- NFS lets many computers use one shared folder over the network, as if it were local.
- AWS EFS is managed NFS: regional, multi-AZ, grows automatically, port 2049, NFSv4.1.
- Create EFS, allow port 2049, install the client tools, mount, and verify with df -h.